

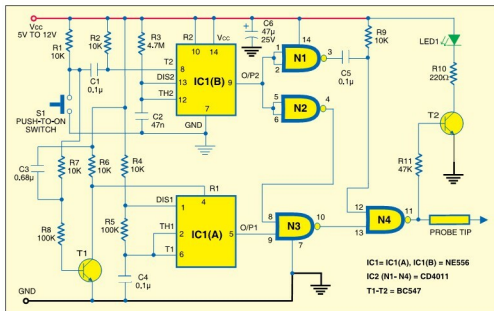


Fig. 1: Digital pulser circuit

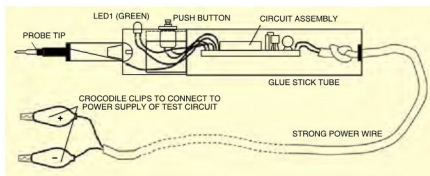


Fig. 2: Assembled unit

goes high for about 1.1R3C2 seconds (where R3=4.7 MΩ and C2=47 nF). The

output of the monostable is inverted by the short-pulse-generating circuit.

LED1 flashes to indicate a single pulse. The single pulse lasts for a short duration as compared to the time period of the monostable IC1(B). Alternatively, you can also see this pulse on an oscilloscope connected between the probe tip and ground. The output of gate N2 is fed to gate N3, which produces the pulse train at the output of gate N4.

Pressing switch S1 enables the astable multivibrator (IC1(A)). Its output remains masked until the output of the monostable (IC1(B)) goes low. When the output of IC1(B) goes low and switch S1 is still kept pressed, the output waveform will be the same as the astable multivibrator's waveform. It can be seen on the probe tip with the help of the oscilloscope. IC1(A) oscillates only when S1 is pressed. Otherwise, the astable remains in reset state.

In short, the digital pulser produces the pulses at the probe tip in the order of single sharp-lapse-train of pulses when switch S1 is pressed.

The entire circuit can be housed in a glue-stick tube as shown in Fig. 2. LED1 is used for pulse indication and incorporated in the probe to make the output visible. ●

AUTOMATIC SPRINKLER CONTROL SYSTEM

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If you have an automatic sprinkler control system, chances are that you are currently using all of its outputs controlling sprinklers. The problem arises when you want to add another sprinkler control solenoid valve. Using this circuit, you can add an expansion module to the controller without upgrading the controller

or running new wiring. The circuit is fitted in place of an existing solenoid and allows the controller to switch on two valves (hence two sprinklers) instead of one.

Designed for use with standard 24V AC controller systems, the circuit uses the 24V output to power a timer which activates one of the valves for, say, 20 minutes. After 20 minutes, the circuit switches this valve 'off' and the

other valve 'on.' The second valve is activated for the remainder of time that the controller is programmed to give for that station.

The 24V AC from the controller is half-wave-rectified by diode D1 and filtered by capacitors C1 and C2. The rectified DC is fed to the timer circuit through resistor R1 and zener diode ZD1 to produce 12 volts to enable the

